

Title - FinTrack: Improving asset returns prediction accuracy through quantile-based deep learning methods.

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Research Question

How can deep learning models be optimized to predict asset returns at various quantile levels and improve the understanding of return distributions and volatility across diverse asset classes?

Summary

The research aims to develop deep learning models for predicting asset returns at different quantile levels (25th, 50th, and 75th percentiles). Models like LSTMs, CNN-LSTM hybrids, and transformers would be used to capture patterns in normalized asset returns adjusted for volatility using risk metrics such as the Sharpe ratio. The focus on quantile-based predictions provides critical insights into return distributions, important for effective portfolio management and financial decision-making.

The research emphasizes leveraging technologies to model a diverse range of asset classes, including stocks, commodities, and cryptocurrencies, with the goal of improving predictive accuracy and robustness. Because of the complexities and volatility of financial markets, this study seeks to identify the most effective architectures for capturing non-linear dependencies, handling multivariate time-series data, and adapting to dynamic market conditions. By bridging gaps in existing predictive methodologies, the project aims to contribute to the advancement of financial analytics, empowering more informed investment strategies and risk management frameworks.

Methodology

- Data Collection and Preprocessing
 - Gather datasets of historical asset returns from sources like Yahoo Finance and Quandle
 - Normalize these returns and adjust for volatility using risk metrics like the Sharpe Ratio
 - Perform exploratory analysis to identify the key features for the model to be trained
- Model Development
 - Design and train deep learning models such as LSTMs, CNN-LSTM hybrids, and transformers to analyze the historical returns adjusted for volatility
 - Explore the performance of hybrid architectures to leverage both temporal (time-based/sequential) and spatial (correlation among different assets)
 - Train simple models at first to ensure an understanding before progressing to more complex architectures

- Training and Validation
 - Split datasets into training, validation, and testing subsets
 - Train models using GPUs to accelerate processing
 - Evaluate models using metrics such as Mean Absolute Error, Mean Squared Error, and Quantile Loss
- Research Collaboration and Ideation
 - Work with team members to analyze relevant research papers, as well as encouraging them to find their own
 - Encourage unique ideas to improve performance, allowing the team to present their ideas
- Implementation
 - Implement the models in real world scenarios to test the practical applicability
 - This includes testing it on various asset classes
 - This also includes testing it over unique time periods in finance (such as the COVID market)
 - Evaluate and optimize models based on performance

Resources Needed

1. Lab Computer // Computing Resources with GPUs - Needed to handle the large datasets of historical financial data and training models with significant epochs
2. Guidance from faculty advisors specializing in machine learning and financial modeling - Important to receive direction and feedback throughout the project

Annotated Bibliography

1. **Paper Title:** *Quantile Deep Learning Models for Multi-Step Ahead Time Series Prediction*
 - a. **Source:** <https://arxiv.org/abs/2411.15674v1>
 - b. **Description:** This paper discusses incorporating quantile regression into models for time series prediction, specifically in the context of highly volatile financial assets like cryptocurrencies. It evaluates hybrid models, including CNN-LSTMs, bidirectional LSTMS, and encoder-decoder LSTMS.
 - c. **Evaluation:** The results indicate the encoder-decoder LSTM outperforms the other models, suggesting it is well-suited for assets with significantly volatile data. It also provides valuable insights into the effectiveness of hybrid models and quantile regression for predicting returns that could be incorporated into our research.
2. **Paper Title:** *Generalized Distribution Prediction for Asset Returns*
 - a. <https://arxiv.org/pdf/2410.23296v1>
 - b. **Description:** This paper introduces a two-stage model that predicts the distribution of returns using quantile-based LSTM networks. The first stages estimate quantiles of normalized returns (mean return / volatility), with the second

integrating broader market data as scaling volatility factors. Then, a kernel density estimation approach transforms quantiles into probability distributions.

- c. **Evaluation:** The model demonstrates generalizability across asset classes, outperforming traditional quantile regression on real data. Two-stage approaches and kernel density estimation are two potential methods for capturing asset return distributions, providing practical insights for generalizability and adaptability across classes.

3. **Paper Title:** *Stock Price Prediction Based on Hybrid CNN-LSTM Model*

- a. https://www.researchgate.net/publication/385730571_Stock_price_prediction_based_on_hybrid_CNN-LSTM_model
- b. **Description:** The study presents a hybrid model with the CNN extracting local temporal features like short-term price changes and volatility clustering, while the LSTM captures long-term dependencies and avoids issues like vanishing gradients.
- c. **Evaluation:** The hybrid model outperforms the standalone CNN and LSTM models, suggesting potential hybrid ideas that could be used for our research.

4. **Paper Title:** *Volatility Forecasting with Hybrid Neural Networks Methods for Risk Parity Investment Strategies*

- a. <https://www.sciencedirect.com/science/article/pii/S095741742300920X>
- b. **Description:** The paper presents a hybrid methodology for forecasting volatility in multi-asset portfolios, combining traditional statistical models with neural networks. The approach employs GARCH models for preprocessing to capture volatility clustering and hybrid RNN architectures (GRU, LSTM, and mixed GRU-LSTM layers) for improved prediction accuracy.
- c. **Evaluation:** Being able to pull volatility data from certain asset types and classes is important to be able to predict asset returns. In this study, the GRU-GARCH hybrid model outperforms other configurations, pointing out potential models that could be used.